PROBLEM NS15

# Moving Porous Wall with Suction

Original statement · NS15.pdf

Asymptotic suction profile | Wall shear and heat flux | Integral balances | Viscous dissipation

### OFFICIAL PROBLEM STATEMENT

Problem 2: Consider a buoyancy-free, steady planar motion occurring in a semi-infinite space filled with a perfect liquid of density $\rho$, viscosity $\mu$, thermal conductivity $k$, and specific heat $c$. The liquid is bounded by an infinitely long porous wall that moves parallel to itself with velocity $U_w$ while removing the liquid with a normal suction velocity $V_w$. Far from the wall, the horizontal velocity component is zero and the pressure is $p = p_\infty$. The ambient value of the temperature $T = T_\infty$ is higher than the value $T_w$ at the wall.

1. Determine the velocity components $v_x(y)$ and $v_y(y)$ and well as the pressure field $p(x, y)$.

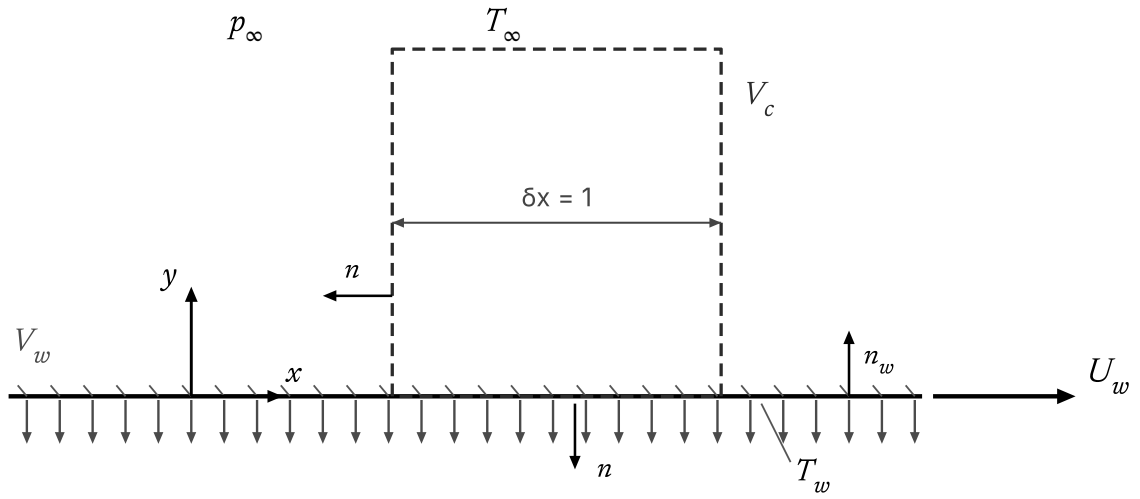
2. Obtain the temperature distribution $T(y)$.

3. Compute the viscous force acting on the wall per unit surface $\tau_w \bar{e}_x = \bar{\bar{\tau}}' \cdot \bar{n}_w$, where $\bar{n}_w$ is the unit vector normal to the wall.

4. Determine the heat flux from the fluid to the wall $q_w = k(\nabla T) \cdot \bar{n}_w$.

5. For an alternative computation procedure for $\tau_w$ and $q_w$, apply the momentum and energy conservation equations to the control volume indicated in the figure. In dealing with the balance equations, evaluate explicitly the contribution of each term, stating clearly the reasons justifying any simplifications introduced.

The sheet is headed "INTRODUCTION TO FLUID MECHANICS, FINAL (Jan 21, 2014)".




Statement figure (redrawn from NS15.pdf, p. 1): porous wall at temperature $T_w$ moving with $U_w$ and sucking with $V_w$, axes $x$, $y$, normals $\bar{n}$ and $\bar{n}_w$, far field $p_\infty$, $T_\infty$ and the fixed control volume $V_c$ of unit length ($\delta x = 1$)

`Source: NS15.pdf (page 1). Notes.pdf, Chapter 4: continuity Eq. 4.40, momentum Eq. 4.41, energy Eq. 4.42, integral balances Eqs. 4.3 and 4.25.`

**Worked solution (4-phase method)**

`PHASE 1`

## Hypotheses, Data and Unknowns

- Steady planar flow of a perfect liquid (constant $\rho, \mu, k, c$), buoyancy-free (mass forces neglected), in the half space $y > 0$ above a porous wall. The fields depend only on the distance $y$ to the wall (the wall is infinitely long and the flow is the same at every $x$).
- The wall at $y = 0$ moves in its own plane with $U_w$ and removes the liquid with the suction velocity $V_w > 0$: $v_y(0) = -V_w$. No chemical reaction and no radiation.
- Far from the wall ($y \to \infty$): $v_x \to 0, p \to p_\infty, T \to T_\infty$, with $T_\infty > T_w$.
- Data: $[\rho,\ \mu,\ k,\ c,\ U_w,\ V_w,\ p_\infty,\ T_\infty,\ T_w]$. Unknowns: $v_x, v_y, p, T, \tau_w$ and $q_w$.
- Notation: $\nu = \mu/\rho, \alpha = k/(\rho c), \mathrm{Pr} = \nu/\alpha = \mu c/k$.

| Symbol | Meaning | SI unit |
|---|---|---|
| $U_w, V_w$ | wall velocity along itself, suction velocity | m/s |

| Symbol | Meaning | SI unit |
|---|---|---|
| $p_\infty, T_\infty, T_w$ | far-field pressure and temperature, wall temperature | Pa, K, K |
| $\tau_w$ | viscous force per unit area on the wall, along $x$ | Pa |
| $q_w$ | heat flux from the liquid to the wall | W/m$^2$ |
| Pr | Prandtl number | 1 |

PHASE 2

## Frames, Field Form and Boundary Conditions

- Cartesian frame $(x, y, z)$ attached to the wall: $x$ along the wall in the direction of $U_w$, $y$ normal to the wall and into the liquid, $z$ out of the plane; the basis is orthonormal and right-handed. The normal $\bar{n}_w = +\bar{e}_y$ points from the wall into the liquid (figure).
- Field form: $\bar{v} = v_x(y)\,\bar{e}_x + v_y(y)\,\bar{e}_y, p = p(x, y), T = T(y)$.
- Control volume $V_c$ of the figure: fixed, of unit length in $x$ ($\delta x = 1$) and unit width, from the wall to a plane $y = Y$ far from it; surfaces: the wall $\Sigma_w$ ($\bar{n} = -\bar{e}_y$), the top $\Sigma_t$ ($\bar{n} = +\bar{e}_y$) and the two vertical faces $\Sigma_1$ ($\bar{n} = -\bar{e}_x$) and $\Sigma_2$ ($\bar{n} = +\bar{e}_x$).

| Location | Condition | Origin |
|---|---|---|
| $y = 0$ | $v_x = U_w, v_y = -V_w, T = T_w$ | moving porous wall, suction |
| $y \to \infty$ | $v_x \to 0, p \to p_\infty, T \to T_\infty$ | liquid at rest far from the wall |

PHASE 3

## Step-by-Step Derivation

### 3.1 Velocity and pressure (item 1)

$$\frac{\partial v_x}{\partial x} + \frac{dv_y}{dy} = \underbrace{\frac{\partial v_x}{\partial x}}_{v_x = v_x(y)} + \frac{dv_y}{dy} = 0 \implies v_y = \text{const} = -V_w \qquad (1)$$

The $y$ component of the momentum equation (no mass force, $dv_y/dy = 0, d^2v_y/dy^2 = 0$) gives $0 = -\partial p/\partial y$, so $p = p(x)$, and the far-field condition $p \to p_\infty$ for every $x$ gives $\partial p/\partial x = 0$:

$$\boxed{v_y(y) = -V_w, \qquad p(x,y) = p_\infty} \tag{2}$$

The $x$ component:

$$\rho\Big(\underbrace{v_x\frac{\partial v_x}{\partial x}}_{\partial_x = 0} + v_y\frac{dv_x}{dy}\Big) = -\underbrace{\frac{\partial p}{\partial x}}_{=0} + \mu\frac{d^2 v_x}{dy^2} \implies \frac{d^2 v_x}{dy^2} + \frac{V_w}{\nu}\frac{dv_x}{dy} = 0 \tag{3}$$

Trying $e^{\lambda y}$: $\lambda(\lambda + V_w/\nu) = 0$, so $v_x = C_1 + C_2 e^{-V_w y/\nu}$. The condition $v_x \to 0$ at infinity gives $C_1 = 0$ (the exponential decays because $V_w > 0$), and $v_x(0) = U_w$ gives $C_2 = U_w$:

$$\boxed{v_x(y) = U_w\, e^{-V_w y/\nu}} \tag{4}$$

### 3.2 Temperature (item 2)

With $v_y = -V_w$, $\partial_x = 0$ and $\phi_v = \mu\,(dv_x/dy)^2$, equation (4.42) reads $-\rho c V_w\, dT/dy = \phi_v + k\, d^2T/dy^2$. From (4), $dv_x/dy = -(V_w U_w/\nu)e^{-V_w y/\nu}$, so

$$\frac{d^2T}{dy^2} + \frac{V_w}{\alpha}\frac{dT}{dy} = -\frac{\mu U_w^2 V_w^2}{k\,\nu^2}\, e^{-2V_w y/\nu} \tag{5}$$

The homogeneous solution is $B_1 + B_2 e^{-V_w y/\alpha}$. A particular solution $A\,e^{-2V_w y/\nu}$ requires (substituting $T_p' = -(2V_w/\nu)A\,e^{-2V_w y/\nu}$ and $T_p'' = (4V_w^2/\nu^2)A\,e^{-2V_w y/\nu}$ and using $\nu/\alpha = \mathrm{Pr}$):

$$A\,(4 - 2\,\mathrm{Pr}) = -\frac{\mu U_w^2}{k} = -\frac{\mathrm{Pr}\,U_w^2}{c} \implies A = \frac{\mathrm{Pr}}{2(\mathrm{Pr}-2)}\frac{U_w^2}{c} \tag{6}$$

for $\mathrm{Pr} \neq 2$. The conditions $T(\infty) = T_\infty$ and $T(0) = T_w$ give $B_1 = T_\infty$ and $B_2 = T_w - T_\infty - A$:

$$\boxed{T(y) = T_\infty + \left(T_w - T_\infty - \frac{\mathrm{Pr}\, U_w^2}{2(\mathrm{Pr}-2)\, c}\right) e^{-\mathrm{Pr}\, V_w y/\nu} + \frac{\mathrm{Pr}\, U_w^2}{2(\mathrm{Pr}-2)\, c}\, e^{-2V_w y/\nu}} \quad (7)$$

since $V_w y/\alpha = \mathrm{Pr}\, V_w y/\nu$.

### 3.3 Viscous force on the wall (item 3)

With $\bar{n}_w = \bar{e}_y$, the vector $\bar{\bar{\tau}}' \cdot \bar{n}_w$ is the viscous force per unit area that the liquid exerts on the wall: $\tau'_{xy}\, \bar{e}_x + \tau'_{yy}\, \bar{e}_y$ with $\tau'_{xy} = \mu\, dv_x/dy$ and $\tau'_{yy} = 2\mu\, dv_y/dy = 0$. At $y = 0$, from (4):

$$\boxed{\tau_w = \mu \frac{dv_x}{dy}\Big|_0 = -\rho\, V_w\, U_w} \quad (8)$$

The liquid pulls the wall back, against its motion.

### 3.4 Heat flux to the wall (item 4)

With $\bar{n}_w = \bar{e}_y$, $q_w = k\, dT/dy\big|_0$. Differentiating (7) at $y = 0$, with $B_2 = T_w - T_\infty - A$:

$$\frac{dT}{dy}\Big|_0 = -\frac{V_w}{\nu}\left(\mathrm{Pr}\, B_2 + 2A\right) \implies q_w = -\rho c V_w \left(B_2 + \frac{2A}{\mathrm{Pr}}\right) = -\rho c V_w \left[T_w - T_\infty - \frac{(\mathrm{Pr}-2)A}{\mathrm{Pr}}\right] \quad (9)$$

using $k/\nu = \rho c/\mathrm{Pr}$ and $(\mathrm{Pr}-2)A/\mathrm{Pr} = U_w^2/(2c)$:

$$\boxed{q_w = \rho c V_w \left[T_\infty - T_w + \frac{U_w^2}{2c}\right] = \frac{1}{2}\rho V_w U_w^2 + \rho c V_w \left(T_\infty - T_w\right)} \quad (10)$$

### 3.5 Alternative computation with the control volume $V_c$ (item 5)

*Momentum, $x$ component* (steady, fixed volume, Notes.pdf Eq. 4.2 in integral form; no mass force):

$$\int_{\Sigma_c} \rho\, v_x\, (\bar{v}\cdot\bar{n})\, d\sigma = -\int_{\Sigma_c} p\, n_x\, d\sigma + \int_{\Sigma_c} (\bar{\bar{\tau}}'\cdot\bar{n})_x\, d\sigma \tag{11}$$

| **Surface** | $\int \rho v_x\, \bar{v}\cdot\bar{n}\, d\sigma$ | $-\int p\, n_x\, d\sigma$ | $\int (\bar{\bar{\tau}}'\cdot\bar{n})_x\, d\sigma$ |
|---|---|---|---|
| $\Sigma_w$ ($\bar{n} = -\bar{e}_y$, $\bar{v}\cdot\bar{n} = V_w$) | $\rho\, U_w V_w$ | $0$ ($n_x = 0$) | $-\tau'_{xy}(0) = -\tau_w$ |
| $\Sigma_t$ ($\bar{n} = +\bar{e}_y$) | $0$ ($v_x \to 0$ far from the wall) | $0$ ($n_x = 0$) | $0$ ($dv_x/dy \to 0$) |
| $\Sigma_1 + \Sigma_2$ | cancel (profile independent of $x$, opposite normals) | cancel ($p$ uniform) | cancel ($\tau'_{xx} = 0$ ) |

Therefore $\rho\, U_w V_w = -\tau_w$, that is $\tau_w = -\rho V_w U_w$, in agreement with (8): the wall must supply the momentum $\rho U_w$ per unit mass to the liquid it sucks, since the liquid at infinity has none.

*Total energy* (Notes.pdf, Eq. 4.3 for a fixed volume, steady, no mass force):

$$\int_{\Sigma_c} \rho \left(e + \frac{|\bar{v}|^2}{2}\right) \bar{v}\cdot\bar{n}\, d\sigma + \int_{\Sigma_c} p\, \bar{v}\cdot\bar{n}\, d\sigma = \int_{\Sigma_c} \bar{v}\cdot\bar{\bar{\tau}}'\cdot\bar{n}\, d\sigma - \int_{\Sigma_c} \bar{q}\cdot\bar{n}\, d\sigma \tag{12}$$

| **Surface** | **Convective energy flux** | **Pressure work** | **Viscous work** $\bar{v}\cdot\bar{\bar{\tau}}'\cdot\bar{n}$ | **Conduction** $\bar{q}\cdot\bar{n}$ |
|---|---|---|---|---|
| $\Sigma_w$ | $\rho\left[cT_w + \frac{1}{2}(U_w^2 + V_w^2)\right] V_w$ | $p_\infty V_w$ | $-U_w\tau_w = \rho V_w U_w^2$ | $k\, dT/dy\big\vert_0 = q_w$ |
| $\Sigma_t$ | $-\rho\left[cT_\infty + \frac{1}{2}V_w^2\right] V_w$ | $-p_\infty V_w$ | $0$ (no shear far away) | $0$ ($dT/dy \to 0$) |
| $\Sigma_1 + \Sigma_2$ | cancel | cancel | cancel | $0$ ($T$ independent of $x$) |

On $\Sigma_w$, $\bar{v}\cdot\bar{n} = V_w$, $\bar{v} = U_w\bar{e}_x - V_w\bar{e}_y$ and $\bar{v}\cdot\bar{\bar{\tau}}'\cdot\bar{n} = -(v_x\tau'_{xy} + v_y\tau'_{yy}) = -U_w\tau_w$; on $\Sigma_t$, $\bar{v} = -V_w\bar{e}_y$ and $T = T_\infty$. The pressure is uniform ($p_\infty$) so its work cancels between $\Sigma_w$ and $\Sigma_t$. Adding the surfaces,

$$\rho V_w\left[c\,(T_w - T_\infty) + \frac{U_w^2}{2}\right] = \rho V_w U_w^2 - q_w \implies \boxed{q_w = \frac{1}{2}\rho V_w U_w^2 + \rho c V_w\,(T_\infty - T_w)} \tag{13}$$

which is (10). The simplifications used are justified by the fact that all the fields are independent of $x$ (lateral faces), that the shear and the temperature gradient decay exponentially (top surface, placed at $Y \gg \nu/V_w, \alpha/V_w$), and that the liquid is at rest and at $p_\infty$ far away.

PHASE 4

### Interpretation, Limits and Dimensional Check

- **Physical meaning.** Suction holds the moving layer of thickness $\nu/V_w$ (momentum) and $\alpha/V_w$ (thermal) against the wall. The wall pays for the momentum of the entrained liquid ((8)), and receives the enthalpy, the kinetic energy and the viscous work ((10)).
- **Limits.** For $U_w = 0$: no shear, no dissipation, $q_w = \rho c V_w (T_\infty - T_w)$ (pure advection-conduction). For $V_w \to 0$ the layer thickens and there is no steady solution. For $T_\infty = T_w$ the heat flux is $\frac{1}{2}\rho V_w U_w^2 > 0$: the wall must remove the kinetic energy of the entrained liquid even when the temperatures are equal.
- **Dimensional check.** $V_w y/\nu$ and $\mathrm{Pr}\, V_w y/\nu$ are dimensionless. $\rho V_w U_w$ has units $\mathrm{N/m^2}$. $\rho V_w U_w^2$ and $\rho c V_w \Delta T$ have units $\mathrm{W/m^2}$.
- **Relation to NS1.** The problem is that of NS1 with the wall moving instead of the free stream: the profiles have the same form, and (10) is (23) of NS1 with $U_\infty \to U_w$.